Dushyant **Shukla**

Software Engineer (Graphics, Game Engine & Tools)

dushyantshukla@outlook.com | dushyant-shukla.github.io/showcase/ | linkedin.com/in/dushyantshukla/

Graphics software engineer specializing in real-time interactive simulation systems, with expertise in real-time rendering, game engine and tools development.

Technical Skills

Programming	C++, C, Lua, C#
Graphics & Physics	Vulkan, OpenGL, GLSL, Unity 3D, NVIDIA PhysX SDK, RenderDoc, NVIDIA Nsight, Blender
Software Development	Protocol Buffers, gRPC, SQLite, Git, CMake, Premake, GitHub Actions, Jira, Confluence, Agile
Core Expertise	Real-time rendering, engine & tools development, GPU/CPU performance profiling & optimization

Professional Experience

Mimic Technologies, Inc. (Surgical Science Sweden AB)
Seattle, WA, USA

Software Developer (Research & Development)
June 2021 - Present

- Developing and maintaining high-performance real-time graphics and simulation systems using C++, proprietary engines, Unity3D, & Nvidia PhysX for high-fidelity robotic surgery training products across multiple platforms.
- Lead engineer for a needle-driving and suturing simulation module, researching soft-body simulation techniques, developing engine-side features, creating scenario behaviors, and collaborating with artists to prepare and integrate art assets.
- Led collaboration with external studios to modernize and optimize a proprietary legacy C++ medical simulation engine, adding support for current OS platforms and enabling renewed product revenue.
- Migrated a proprietary C++ real-time surgical simulation engine from x86 to x64 architecture and implemented a Premake-based build system, improving performance and maintainability, and developer workflows.
- Led development of a low-latency real-time data pipeline capturing high-resolution surgical performance telemetry (C++, Protocol Buffers, reliable-UDP, SQLite) to support personalized machine-learning driven feedback. Owned release planning and workflows.
- Designed and implemented a database migration tool (C++, Dear ImGui) to automate complex schema migrations, replacing an error-prone multi-step manual workflow and ensuring successful contract fulfillment for a key client.
- Supported developer workflows by providing documentation, engaging in architecture discussions, and conducting code reviews.
- Delivered technical presentations during internal developer seminars to educate the team and promote knowledge sharing.

SenecaGlobal IT Services Private Limited
Hyderabad, TG, India

Technical Analyst (Open Source Technologies)
September 2015 - July 2019

- Spearheaded the development of a distributed digital asset management workflow using AWS, Spring Boot, and Dropbox APIs, reducing customer processing time and service costs by 50% and enabling a new revenue stream for the client.
- Collaborated with geographically distributed teams to provide service & support, ensuring rapid issue resolution.
- Provided technical guidance and mentorship to software engineers, contributing to skill development and team effectiveness.

Education

M.S. in Computer Science, DigiPen Institute of Technology — *April 2021* — *Redmond, WA, USA*

B.Tech. in Computer Science & Engineering, Dr. A.P.J Abdul Kalam Technical University — *June 2015* — *India*

Projects

Cassini Engine (R&D Custom Game Engine Project)

Exploring advanced graphics techniques & engine architectures (C++, Vulkan) | Homepage
June 2024 - Present

- Implemented a physically based rendering (PBR) pipeline using the metallic-roughness workflow with glTF asset support.
- Added bindless rendering to reduce draw-call overhead and improve GPU resource access efficiency.
- Developed asynchronous resource loading to prevent stalls and improve runtime performance.

Age of Empyrion, Advanced Game Project (DigiPen Institute of Technology)
Redmond, WA, USA

Engine & Gameplay Programmer(C++) | 4 member team | Homepage
January 2020 - April 2020

- Engineered a type-safe, cache-aware Entity Component System (ECS) leveraging C++ templates and data-oriented design to manage large-scale game state efficiently.
- Built a C++ template-based publish–subscribe event system to support decoupled, scalable inter-system communication.
- Implemented a modular inventory and ability system using virtual inheritance, enabling flexible composition of player skills.